

Support vector machines II: kernels and soft margin

Feature maps, the kernel trick, valid kernels, RBF and γ , slack variables, C and the hinge loss.

THE QUESTION

How does a maximum-margin classifier draw a curved boundary, and what should it do when no boundary separates the classes?

THE INTUITION

Map each point to richer features $\varphi(x)$, where a flat boundary can separate them. The dual needs only dot products, so a kernel $K(x, z) = \varphi(x)^T \varphi(z)$ supplies them without building φ . Slack ξ_i lets a few points into the street, at a price C each.

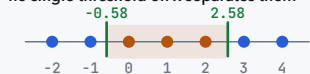
KEY TAKEAWAY

Replace $x_i^T x_j$ by $K(x_i, x_j)$ in the dual and bound $0 \leq \alpha_i \leq C$. Predict $\text{sign}(\sum_j \alpha_j y_j K(x_j, x) + b)$. The RBF width γ and the price C set the bias and the variance.

$\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^D \rightarrow$ dual needs $x_i \cdot x_j$ only $\rightarrow K(x_i, x_j) = \varphi(x_i) \cdot \varphi(x_j) \rightarrow$ Gram matrix PSD $\rightarrow 0 \leq \alpha_i \leq C \rightarrow$ hinge + $\|w\|^2/2C$

1 A curve in x is a line in (x, x^2)

no single threshold on x separates them



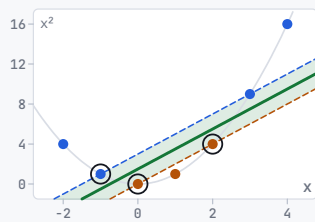
$\varphi(x) = (x, x^2)$; a line there:

$$(4/3)x - (2/3)x^2 + 1 = 0$$

back in 1D: $(x-1)^2 = 5/2$

$$x = 1 \pm \sqrt{2.5} = -0.58 \text{ or } 2.58$$

7 points, labels from the kernel notebook



Class +1 sits between two -1 groups, so no threshold works. Map $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^D$ ($d=1, D=2$) and fit the maximum-margin line there: $w = (4/3, -2/3)$, $b = 1$ (the notebook prints 1.3, -0.7, 1.0), support vectors $x = -1, 0, 2$ ($\alpha = 1.11, 1, 0.11$). Linear in the parameters, like basis functions (the linear regression II sheet).

2 The dual needs dot products, never φ itself

features D of $(1+xz)^q$ for x in $\mathbb{R}^d$, against d multiplications for K

d	q	D	d	q	D
2	2	6	100	2	5,151
2	3	10	100	3	176,851
10	2	66	1000	3	167,668,501

After the map the dual (the SVM I sheet) reads

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \varphi(x_i)^T \varphi(x_j).$$

Building every $\varphi(x)$ and taking D -term dot products is slow when $D \gg d$. The primal holds $w \in \mathbb{R}^D$ itself, so only the dual allows the swap to K . The slides' $(1+x^T z)^3$ on $\mathbb{R}^2$ has 10 features, weighted by $\sqrt{3}$ and $\sqrt{6}$.

3 The kernel trick, checked by hand

$x = (1, 2), z = (3, -1): x \cdot z = 1$, so $(1+x \cdot z)^2 = 4$

	1	$\sqrt{2} x_1$	$\sqrt{2} x_2$	x_1^2	x_2^2	$\sqrt{2} x_1 x_2$
$\varphi(x)$	1	1.41	2.83	1	4	2.83
$\varphi(z)$	1	4.24	-1.41	9	1	-4.24
product	1	6	-4	9	4	-12
						$\Sigma = 4$

1D, as on the slides: $\varphi(x) = (\sqrt{2} x, x^2)$, $K = (1+xz)^2 - 1$

$$\varphi(-2) \cdot \varphi(-1) = (-2\sqrt{2})(-\sqrt{2}) + 4 \cdot 1 = 8 = (1+2)^2 - 1$$

For $x \in \mathbb{R}^2$, $(1+x^T z)^2 = \varphi(x)^T \varphi(z)$ with the six features in the header. Six products sum to 4, and one dot product plus a square gives the same 4. Over 200 random pairs the two agree to 4×10^{-15} . K is a similarity computed in the original space.

4 A valid kernel gives a PSD Gram matrix

$K = (1+x_i x_j)^2 - 1, x = -2$ to 4

	-2	-1	0	1	2	3	4
-2	24	8	0	0	8	24	48
-1	8	3	-1	0	3	8	8
0	0	0	0	0	0	0	0
1	0	-1	0	3	8	15	24
2	8	0	0	8	24	48	80
3	24	3	0	15	48	99	168
4	48	8	0	24	80	168	288

outlined: 3, the slide has 1

eigenvalues of the 7 x 7 matrix

$(1+xz)^2 - 1$, the slide's numbers

418.5, 22.5, 0, 0, 0, 0, 0

rank 2 = D: PSD

$(1+xz)^2$, the slide's formula

421.7, 22.9, 3.4, 0, 0, 0, 0

rank 3 = D: PSD

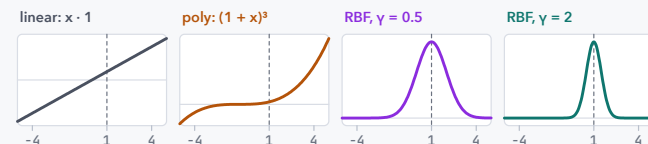
$(x-z)^2$, a distance

65, 0, 0, 0, 0, -9, -56

negative: not a kernel

The Gram matrix $K_{ij} = K(x_i, x_j)$ of any valid kernel is symmetric and positive semi-definite: $v^T K v = \|\sum_i v_i \varphi(x_i)\|^2 \geq 0$. Its rank is at most D . On the slides' seven points the kernel behind the numbers is $(1+xz)^2 - 1$, so the row for $x = 0$ is all zeros. A squared distance fails the test.

5 Three kernels in common use



Linear $x^T z$; polynomial $(c+x^T z)^q$ (constant c , degree q); RBF or Gaussian $\exp(-\gamma \|x-z\|^2)$ with $\gamma = 1/2\sigma^2$. Plotted against x for a fixed point 1: the polynomial grows without bound, the RBF is 1 at the point and fades, faster for larger γ . $(x^T z)^2$ maps $\mathbb{R}^2$ to $(x_1^2, \sqrt{2}x_1x_2, x_2^2)$, which separates $(0, \pm 1)$ from $(\pm 1, 0)$.

6 RBF has infinitely many features

1D: $K(x, z) = e^{-(\gamma x^2 - 2\gamma xz + \gamma z^2)} = e^{-(\gamma x^2)} e^{(2\gamma xz)} e^{-(\gamma z^2)}$

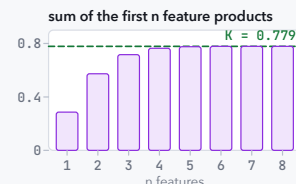
$$e^{(2\gamma xz)} = \sum_k (2\gamma xz)^k / k!, k = 0, 1, 2, \dots$$

$$\varphi_k(x) = e^{-(\gamma x^2)} \sqrt{(2\gamma)^k / k!} x^k$$

one feature per k : infinitely many

$$x = 1, z = 0.5, \gamma = 1: K = 0.779$$

$$x = 2, z = -1.5 \text{ needs 23 features for 1\%}$$

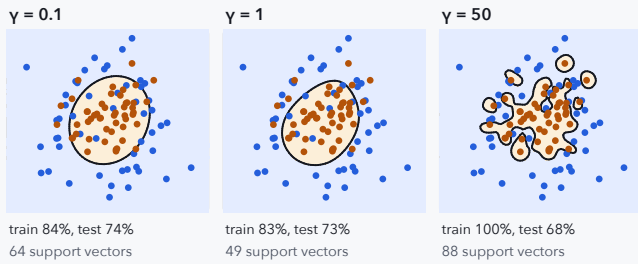


The middle factor holds the 1D dot product xz , and its Taylor series never stops, so φ has one coordinate per power k . Truncated, the sum creeps up to the exact 0.779: 0.29, 0.57, 0.72, 0.76 with 1 to 4 features. The kernel gets it exactly from one exponential.

RBF width, slack and the hinge loss

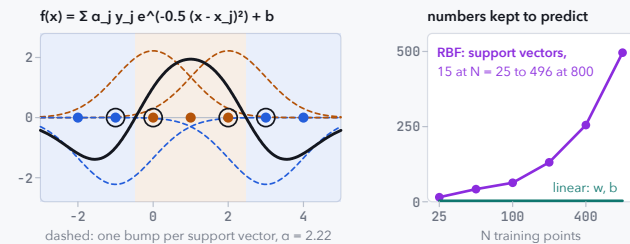
What γ does, why RBF is non-parametric, slack variables and C , and the same SVM written as loss plus penalty.

7 γ sets how far one point reaches



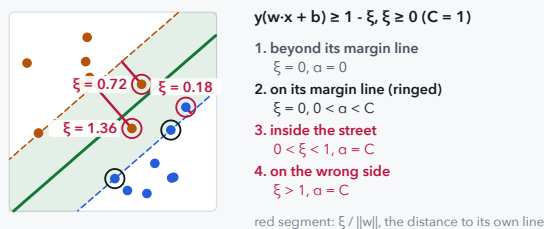
Two noisy rings, 100 points, $C = 10$, as in the kernel notebook (our seed; tested on 2000 fresh points). Small γ : wide bumps, a smooth boundary. Large γ : each point its own island; training accuracy hits 100% and test accuracy falls. $\gamma = 0.01$ is nearly linear (65%; a linear SVM gets 61%). Pick γ and C by cross-validation.

8 RBF adds one bump per support vector



$\hat{y}(x) = \text{sign}\left(\sum_j \alpha_j y_j \exp\left(-\gamma \|x - x_j\|^2\right) + b\right)$. On the seven 1D points ($\gamma = 0.5$): four support vectors, all $\alpha = 2.22$, $b = -0.15$. Linear and polynomial kernels keep a fixed form (w, b) : parametric. RBF must keep its support vectors, and on the rings their number grows with N (15 to 496): non-parametric, like KNN (the KNN sheet).

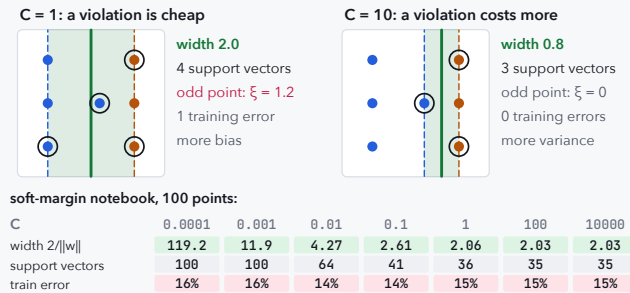
9 Slack lets points into the street



$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

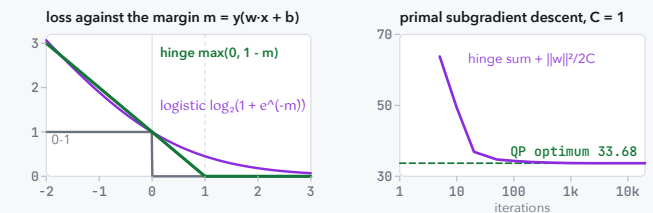
The dual changes only by a box: $0 \leq \alpha_i \leq C$. Any point with $\xi_i > 0$ has $\alpha_i = C$. Here two points sit inside ($\xi = 0.18, 0.72$), one is on the wrong side ($\xi = 1.36$), and two free support vectors ($\alpha = 0.63, 0.37$) fix b .

10 C trades margin width for violations



$C \rightarrow 0$ leaves $\min \frac{1}{2} \|w\|^2$: a wide street that ignores the ξ_i (more training error, more bias). $C \rightarrow \infty$ forbids slack: the hard margin, sensitive to single points (more variance, the bias and variance sheet). On the notebook's blobs the street narrows and support vectors fall from 100 to 35 as C grows. Training error barely moves: it is set by the overlap. $C = 100$ and 10^4 give the same line.

11 The slack is a hinge loss



At the optimum $\xi_i = \max[0, 1 - y_i(w^T x_i + b)]$. Dividing by C :

$$\min_{w, b} \sum_i \max[0, 1 - y_i(w^T x_i + b)] + \frac{1}{2C} \|w\|^2.$$

Hinge plus a convex penalty is convex. The hinge has no derivative at $m = 1$, but subgradients work: 20000 steps reach the QP optimum, $w = (0.176, -0.956)$, $b = 2.47$. Logistic loss (the logistic regression sheet) is never zero, so every point pulls; the hinge ignores points beyond the margin.

12 Pause and think

A support vector with $\xi = 1.5$? Misclassified: $\xi > 1$.

Features of $(1 + x^T z)^2$ on $\mathbb{R}^2$? Six. Train 100%, test 68%? γ too large: lower it.

Hard margin in this form? $C \rightarrow \infty$.

Is $(x - z)^2$ a kernel? No: an eigenvalue of -56 . More than two classes? One vs all (largest $w^T x + b$) or one vs one (majority vote).

$$K(x, z) = \varphi(x)^T \varphi(z) \cdot \text{Gram PSD} \cdot \text{RBF } e^{-\gamma \|x - z\|^2} \cdot 0 \leq \alpha_i \leq C \cdot \text{hinge} + \|w\|^2 / 2C \cdot \text{sign}\left(\sum_j \alpha_j y_j K(x_j, x) + b\right)$$

Before: the support vector machines I sheet (maximum margin); next: the naive Bayes and Bayesian networks sheet. Source: lecture slides "Support vector machines" (kernels), "SVM soft margin classification" and "SVM multi-class classification" with their handwritten notes, and the SVM kernel, kernel-understanding and soft-margin notebooks (nipunbatra.github.io/ml-teaching); solved here with NumPy (SMO on the dual). Corrected: the Gram-matrix slide says $(1 + x_i x_j)^2$ but its numbers are $(1 + x_i x_j)^2 - 1$, whose $K(-1, -1)$ is 3, not 1; the soft-margin dual on the slide lacks the $\frac{1}{2}$ and says minimise for maximise; the notebook's hand-written SVM averages the hinge, which is C/N in this form.